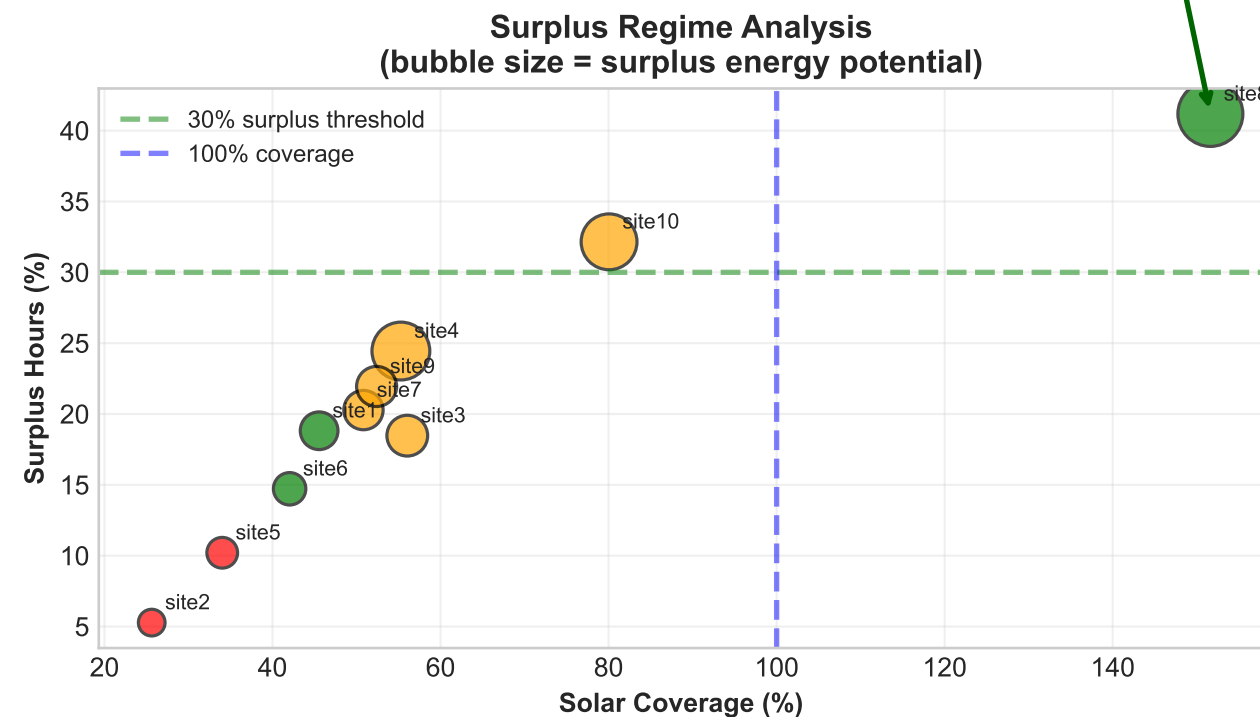
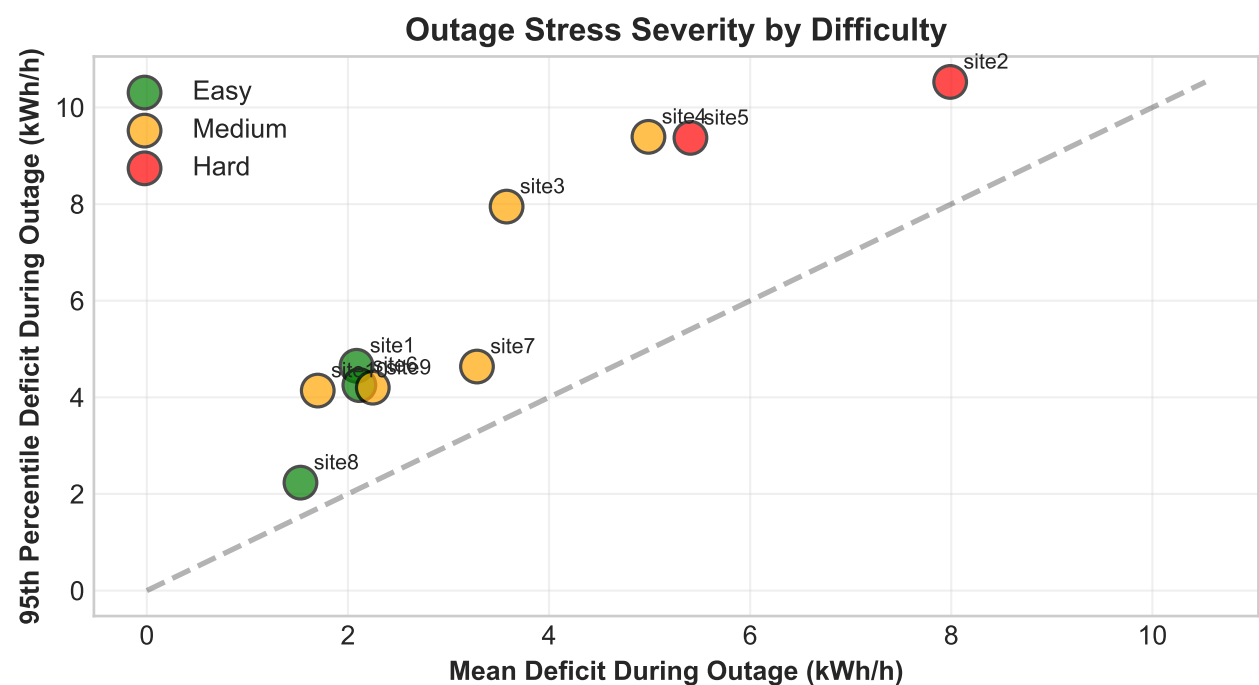
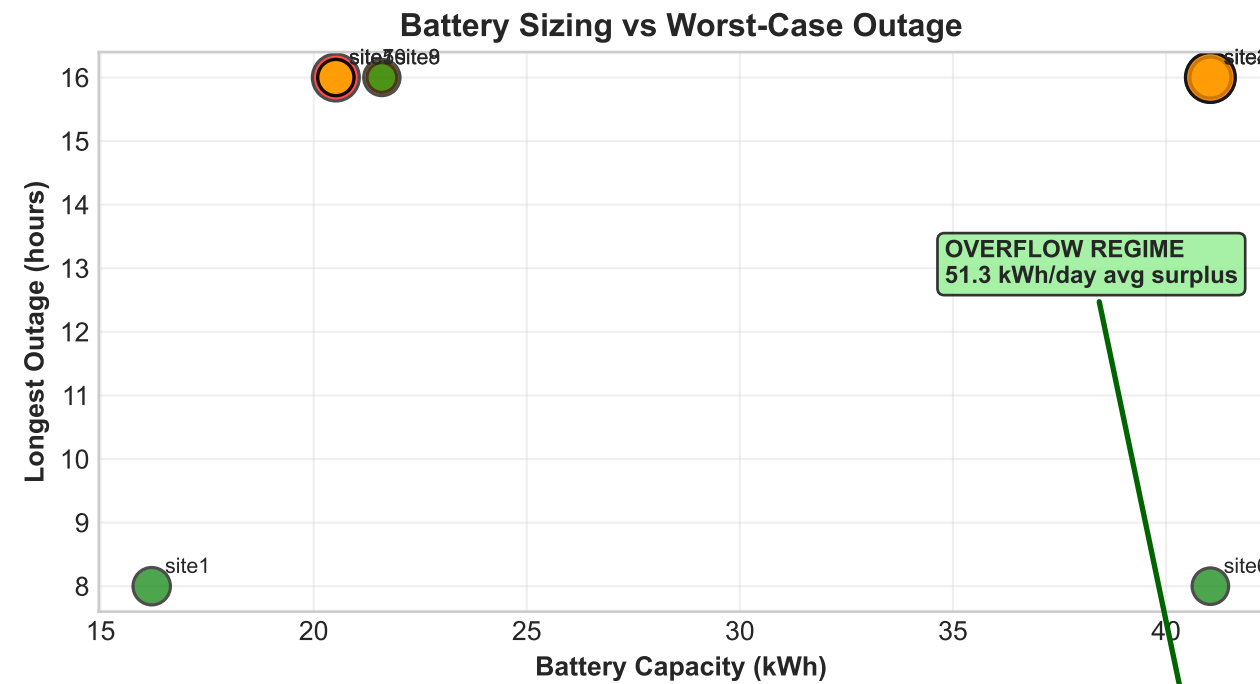
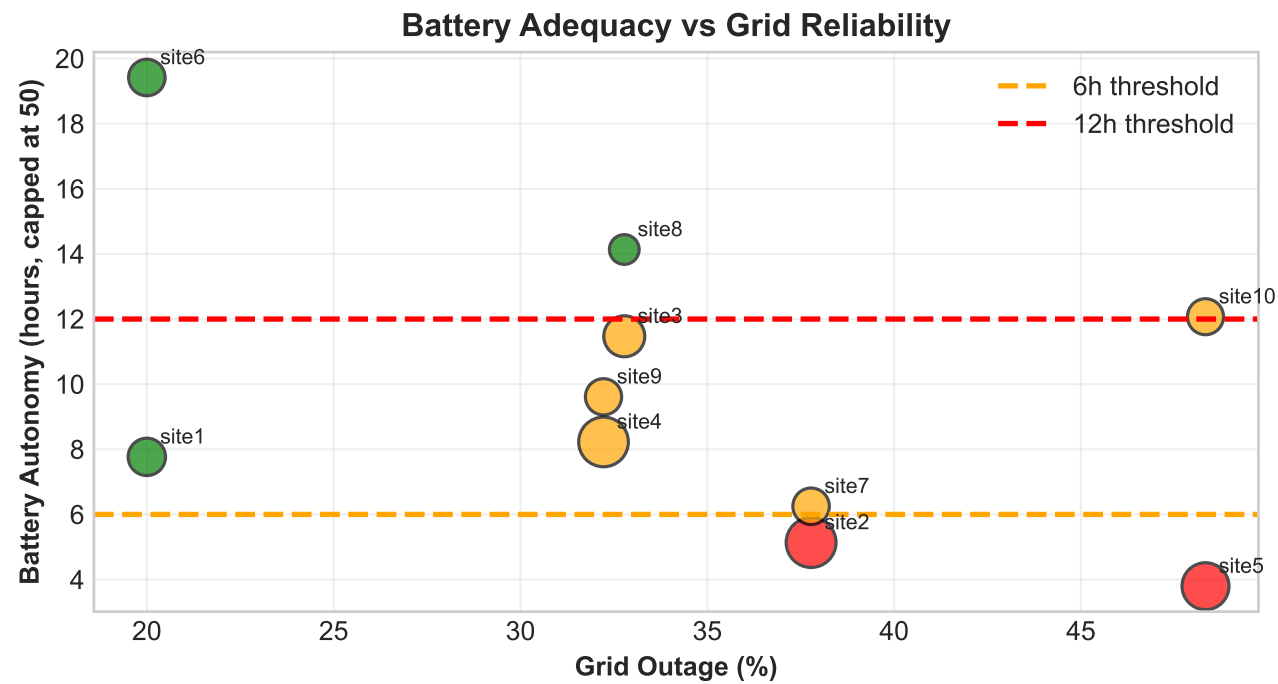


Figure 4: RL Stress Analysis - Battery Adequacy, Outage Severity & Surplus Regime



RL STRESS ANALYSIS - SCARCITY & SURPLUS REGIMES:			
SCARCITY REGIME (Battery Inadequacy):		SURPLUS REGIME (Overflow Potential):	
site5 :	3.8h / 16h = 24%	site8 :	41.2% hrs surplus, 51.3 kWh/day avg surplus
site2 :	5.1h / 16h = 32%	site10:	32.2% hrs surplus, 35.8 kWh/day avg surplus
site7 :	6.2h / 16h = 39%		
DEFICIT STRESS (During Outages):		INVENTORY ANALOGY MAPPING:	
site2 :	7.99 kWh/h mean (p95: 10.53)	Scarcity Regime	↔ Stockout risk (diesel needed)
site5 :	5.41 kWh/h mean (p95: 9.37)	Surplus Regime	↔ Overflow potential (curtailment depends on SoC)
KEY RL INSIGHTS:			
• Battery sizing inadequate for worst-case (diesel CRITICAL for most sites)			
• Site8 tests overflow regime handling (battery saturation + surplus energy management)			
• Policy must handle BOTH scarcity and surplus regimes			
• Inventory analogy: stockout (deficit) + overflow (surplus) + capacity constraints			